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Exp.No-4(B)

Title- TRAVELLING SALESPERSON PROBLEM

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In [ ]: def tsp_dp(distances):
    n, full = len(distances), (1 << len(distances)) - 1
    dp = [[float('inf')] * n for _ in range(1 << n)]
    dp[1][0] = 0

    for mask in range(1 << n):
        for i in range(n):
            if mask & (1 << i):
                for j in range(n):
                    if mask & (1 << j) and i != j:
                        dp[mask][i] = min(dp[mask][i],
                                           dp[mask ^ (1 << i)][j] + distances[j][i])

    return min(dp[full][i] + distances[i][0] for i in range(1, n))

n = int(input("Enter number of cities: "))
print("Enter the cost matrix (enter each row separated by spaces):")
distances = [list(map(int, input().split())) for _ in range(n)]

print("=====")
print("minimum cost of TSP:", tsp_dp(distances))
print("=====")
```

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In [ ]: Enter number of cities: 4
Enter the cost matrix (enter each row separated by spaces):
0 30 4 6
30 0 10 5
4 10 0 20
6 5 20 0
=====
minimum cost of TSP: 25
=====
```